

- 4**
- (a)** two sine waves in antiphase (one dotted, one solid line) **B1**
 4 antinodes and 5 nodes shown **B1**
 - (b)** a second sine wave of same wavelength on same axis **B1**
 separated by a quarter of the wavelength **B1**
 - (c)** plane parallel wavefronts before the opening **B1**
 circular wavefronts showing diffraction after the opening with
 same wavelength and **B1**
 greater than the gap width **B1**

5 (a)